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MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATION – MARCH 2026
Fourth Semester – Discipline Specific Core Course (DSC)
MG4DSCZGY202 – FUNCTIONAL ZOOLOGY
(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum mark :50**

Answer Scheme

Remember (K), Understand (U), Apply (A), Analyze (An), Evaluate (E), Create (C), Skill (S), Interest (I), and Appreciation (Ap).

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A
Multiple Choice Questions
Answer all questions. Each question carries 1 mark.

1. _____ catalyses the conversion of fibrinogen to fibrin. [**K]/ [CO1]**
Ans: a) Thrombin

2. HIV, the virus that causes AIDS, primarily infects and destroys **[U]/[CO4]**
Ans: c) CD4-positive T-cells

3. Which class of immunoglobulin is predominant in mucosal secretions? [**K]/ [CO4]**
Ans: b) IgA

4. Addison's disease is due to hyposecretion of _____. [**K]/ [CO3]**
Ans: a) Cortisol and aldosterone

5. Renin is secreted by: **[U]/[CO1]**
Ans: b) Juxtaglomerular cells

6. Name the disease caused by intake of low quality and quantity of dietary proteins. **[U]/[CO1**]
Ans: b) Kwashiorkor

7. Hormones act by binding with_____. [**K]/ [CO3]**
Ans: d) Receptors

8. The immune cells responsible for antigen presentation include **[U]/[CO4**]
Ans: a) Dendritic cells

9. A major advantage of recombinant vaccines is **[U]/[CO4**]
Ans: c**)** Better safety profile

10. The main feature that distinguishes acquired immunity from innate immunity is: [**K]/ [CO4]**
Ans: c) Immunological memory

11. The action potential spreads to the interior of muscle fibre through: [**K]/ [CO2]**
Ans: b) T-tubule

12. Fight or flight response involves_____.**[U]/[CO3**]
Ans: b) Adrenal medulla hormones

Part B
Short Answer Questions
Answer any 6 questions. Each question carries 3 marks.

13. Explain briefly hybridoma technology. [U]/ [CO4]

Hybridoma technology is used to produce monoclonal antibodies. It involves the fusion of B-lymphocytes with myeloma cells to form hybridoma cells. These cells can divide indefinitely and produce specific antibodies in large quantities for medical and research purposes (**Definition: 1 mark; Method: 1 mark; Function: 1 mark)**

14. Differentiate between conduction in myelinated and non-myelinated nerve fibers. [U]/ [CO2]

Myelinated fibres: Have myelin sheath; conduction is saltatory (jumping) between nodes of Ranvier; faster transmission. (**Any 3 points: 1.5 marks**). **Non-myelinated fibres:** Lack myelin sheath; conduction is continuous along the membrane; slower transmission. (**Any 3 points: 1.5 marks**)

15. State the role of bone marrow in immunity. [K]/[CO4]

Bone marrow as a primary lymphoid organ responsible for the production of all blood cells; Formation and maturation of B-lymphocytes, enabling them to produce antibodies; Production of precursors of T-cells and other immune cells. (**Any 3 points: 3 marks)**

16. What is meant by Hamburger Phenomenon? [K]/[CO1]

The Hamburger phenomenon /chloride shift refers to the exchange of bicarbonate ions and chloride ions across the red blood cell membrane during carbon dioxide transport. This process maintains ionic balance and allows more carbon dioxide to be carried in the blood. It is essential for proper respiration and gas exchange. (**Definition: 1 Mark; Mechanism: 1 mark; Significance: 1 Mark**)

17. Differentiate between axon and dendrite. [K]/[CO2]

Axon: Long, cylindrical projection, usually without Nissl bodies, conducts impulses away from the cell body. **Dendrite:** Short, tapering and highly branched projections with Nissl bodies present, conduct impulses towards the cell body. (**Any 3 points: 3 marks)**

18. Explain the role of TRH and its regulation. [U]/[CO3]

TRH (Thyrotropin Releasing Hormone) is secreted by the hypothalamus and stimulates the anterior pituitary to release TSH (Thyroid Stimulating Hormone). TSH in turn stimulates the thyroid gland to produce thyroid hormones (T_3 and T_4), which regulate metabolism. The secretion of TRH is controlled by a negative feedback mechanism, where high levels of thyroid hormones inhibit the release of TRH and TSH, while low levels stimulate their secretion. (**Role of TRH: 1.5 marks; Regulation: 1.5 marks)**

19. Explain how the structure of the renal corpuscle is suited for filtration. [U]/[CO1]

The renal corpuscle is well adapted for filtration due to its specialized structure. It consists of a glomerulus, a network of capillaries enclosed within Bowman's capsule. The glomerular capillaries have thin, porous walls (fenestrations) that allow easy filtration of fluids. The filtration membrane (capillary wall, basement membrane and podocytes) is selectively permeable, allowing small molecules to pass while preventing proteins and cells. High blood pressure in the glomerulus facilitates ultrafiltration. (**Structure: 1.5 marks; Function: 1.5 marks**) (***Liberal valuation***)

20. What is troponin? [K]/[CO2]

Troponin is a regulatory globular protein complex, with 3 subunits (Troponin T, C and I); found in skeletal and cardiac muscle fibers. It plays a key role in muscle contraction by binding to calcium ions and helping control the interaction between actin and myosin filaments. **(Any 3 points: 3 marks)**

Part C
Short Essay Questions
Answer any 4 questions. Each question carries 5 marks.

21. Compare autoantigens and alloantigens with suitable examples. [K]/[CO4]

Autoantigens are self-antigens present on an individual's own cells. Normally, the immune system does not react to them, but in autoimmune diseases, they may trigger an immune response. **Example:** Antigens involved in rheumatoid arthritis or systemic lupus erythematosus (SLE). (**Definition: 1.5 marks; Example: 1 mark**). **Alloantigens** are antigens from another individual of the same species. They can trigger an immune response when transferred from one person to another. **Example:** ABO blood group antigens during blood transfusion / tissue antigens in organ transplantation. (**Definition: 1.5 marks; Example: 1 mark**).

22. Describe different types of WBCs. [U]/[CO1]

White blood cells (WBCs) are classified into two main types: granulocytes and agranulocytes based on the presence of granules. **Granulocytes**: 1) **Neutrophils:** Most abundant; have a multi-lobed nucleus; perform phagocytosis of bacteria. 2)**Eosinophils:** Have a bilobed nucleus with red granules; involved in allergic reactions and parasitic infections**.** 3)**Basophils:** Contain dark blue granules**;** release histamine and heparin during inflammation. **Agranulocytes**: 4) **Lymphocytes:** Have a large round nucleus; include B-cells and T-cells; responsible for immune responses. 5)

Monocytes: Largest WBCs with kidney-shaped nucleus; differentiate into macrophages and perform phagocytosis. (**Description of Granulocytes: 3 marks; Description of Agranulocytes: 2 marks)**

23. Describe mechanism of antigen-antibody reaction. [U]/[CO4]

The antigen–antibody reaction is a specific immune response in which antibodies bind to their corresponding antigens to neutralize or eliminate them. Antibodies recognize antigens through specific binding sites (paratopes) that fit the antigen's epitopes like a lock and key. When binding occurs, it forms an antigen–antibody complex**. Agglutination:** Clumping of cells or particles; **Precipitation:** Formation of insoluble complexes from soluble antigen. **Neutralization:** Blocking the harmful effects of toxins or pathogens**. Complement activation:** Leads to lysis of pathogens. **Opsonization:** Enhances phagocytosis by marking pathogens for destruction. (**Definition: 1 mark; Mechanism: 1 mark; Reactions: 3 marks).**

24. Outline the causes and consequences of hypoxia and hypercapnia. **[U]/[CO1]**
Hypoxia and hypercapnia are disorders of impaired gas exchange in the body. Hypoxia is mainly caused by lung diseases, high altitude, anaemia, heart failure and carbon monoxide poisoning, while hypercapnia results from hypoventilation due to chronic obstructive pulmonary disease, respiratory depression from drugs, neuromuscular disorders, or airway blockage. The consequences of hypoxia include fatigue, cyanosis, organ dysfunction, brain damage, and death in severe cases. Hypercapnia leads to respiratory acidosis, headaches, confusion, drowsiness and in extreme situations, coma and respiratory failure. (**Causes of hypoxia and hypercapnia: 2 marks; Effects of hypoxia and hypercapnia: 3 marks**)

25. Give an account on rigor mortis. [U]/[CO2]

Rigor mortis is the stiffening of muscles after death due to biochemical changes in muscle fibers. After death, respiration stops and ATP production ceases. ATP is necessary for muscle relaxation, so in its absence, the actin and myosin filaments remain locked, causing muscle stiffness. Rigor mortis usually begins within 2 to 4 hours after death, first appearing in the small muscles of the face and jaw, and then spreads to other parts of the body. It becomes fully developed in about 12 hours and gradually disappears after 24 to 48 hours due to muscle protein breakdown. This process is important in forensic science to estimate the time of death. (**Definition: 2 marks; Cause: 2 marks; Importance: 1 mark)**

26. Write an account on neural disorders in man. [K]/[CO2]

Neural disorders affect the normal functioning of the nervous system. Structural, biochemical or electrical abnormalities in the brain, spinal cord or other nerves can result in wide range of symptoms. Common disorders are: **Dyslexia:** A learning disorder characterized by difficulty in reading, writing and spelling**,** despite normal intelligence. **Parkinson's disease:** A degenerative disorder causing tremors, muscle rigidity, slow movements and poor balance. **Epilepsy:** A neurological disorder marked by recurrent seizures, sudden loss of consciousness and uncontrolled muscle movements. **Alzheimer's disease:** A progressive brain disorder leading to memory loss, confusion, difficulty in thinking and behavioral changes**. (Definition: 1 mark; Neural Disorders with symptoms: 4 marks)**